

A Naimark counterexample of continuum norm density

MathlibAnnex.CStarAlgebra.CAR.existsNaimarkCounterexampleOfDensity_continuum

theorem

Gives a C*-algebra of norm density $\mathfrak{c}$ whose nonzero irreducible representations form one unitary-equivalence class, without an identification with the compact operators.

Statement

There exist a nonzero complex C*-algebra B , not required to be unital, a complex Hilbert space K , and a nonzero irreducible $*$ -representation $\sigma : B \rightarrow B(K)$ with the following properties. Every nonzero irreducible $*$ -representation of B is unitarily equivalent to σ . The representation σ is not both injective and surjective onto $\mathcal{K}(K)$, the full algebra of compact operators on K . Finally,

$$\text{dens}_{\|\cdot\|}(B) = \mathfrak{c}, \quad \mathfrak{c} = 2^{\aleph_0}.$$

Here density refers to the norm topology. None of the representations is required in advance to preserve an identity.

Assumptions

This is an existence theorem with no CH hypothesis. Its statement allows nonunital C*-algebras and places no requirement on the existence of a faithful representation on a separable Hilbert space.

Conclusion

One may take $B = T = C^*(\pi(A) \cup \{L_i : i \in I\})$, where A is CAR, π is the selected GNS direct sum, and L_i are the unitaries from the fixed homogeneity and shell construction. Take $K = \mathcal{H}$ and $\sigma = \rho_0$, the inclusion of this concrete algebra in $B(\mathcal{H})$. Thus the constructed example is unital, even though unitality is not required by the existence statement.

The fixed example also admits a faithful representation on a separable Hilbert space, but that is not a condition in this existence statement and is not asserted for every C*-algebra satisfying it.

Proof route

Use the constructed algebra T and its inclusion representation ρ_0 . The classification theorem gives the single unitary-equivalence class, the compact-operator exclusion applies to ρ_0 , and the density theorem gives the required value $\mathfrak{c}$.

▼ Proof steps

1. Take the algebra T , its selected GNS direct sum $\mathcal{H}$, and the inclusion ρ_0 . The representation-classification theorem says that ρ_0 is nonzero irreducible and that, for any nonzero irreducible $*$ -representation $\rho : T \rightarrow B(H)$, a unitary $U : \mathcal{H} \rightarrow H$ satisfies $U\rho_0(t) = \rho(t)U$. This statement does not require $\rho(1) = I_H$ as an initial assumption.
2. The compact-operator exclusion rules out every injective representation of T onto the full algebra of compact operators, in particular the specified inclusion ρ_0 . The density theorem gives $\text{dens}_{\|\cdot\|}(T) = \mathfrak{c}$. These are precisely the other conditions of the existential predicate in the Lean statement.

Main citations

- The stated existence or structural result —
`MathlibAnnex.CStarAlgebra.CAR.existsNaimarkCounterexampleOfDensity_continuum`
- The exact counterexample and density conditions —
`MathlibAnnex.Analysis.CStarAlgebra.ExistsNaimarkCounterexampleOfDensity`
- Unitary equivalence without an initial identity-preservation assumption —
`MathlibAnnex.CStarAlgebra.CAR.isUniqueIrreducibleModelAmongNonUnital_shellFamilyInclusion`
- The same equivalence-class condition for the nonunital-algebra API —
`MathlibAnnex.Analysis.CStarAlgebra.Representation.IsSingletonIrreducibleModelAmongNonUnital.isSingletonIrreducibleModel_toNonUnitalStarAlgHom`
- Exclusion of an injective representation onto the compact operators —
`MathlibAnnex.CStarAlgebra.CAR.not_isCompactOperatorModel_shellFamilyTarget`
- Exact norm density of the constructed algebra —
`MathlibAnnex.CStarAlgebra.CAR.hasDensityCharacter_atomicCounterexampleAlgebra`

Supporting route explanation

Lean source signature (exact)

```
theorem existsNaimarkCounterexampleOfDensity_continuum :
  ExistsNaimarkCounterexampleOfDensity (Cardinal.continuum : Cardinal.{0})
```

In the source	Mathematical meaning
<code>(Cardinal.continuum : Cardinal.{0})</code>	The cardinal $\mathfrak{c} = 2^{\aleph_0}$ in the universe of Type 0 types. The colon specifies a type and <code>.{0}</code> specifies a universe level, not cardinal zero or countability.
<code>ExistsNaimarkCounterexampleOfDensity</code>	The linked exact existential predicate chooses one nonzero possibly nonunital complex C^* -algebra B , its spectral order, one complete complex Hilbert space K , and one star representation $\sigma : B \rightarrow B(K)$ in the indicated universe. Source letters A, H, π in that definition denote B, K, σ here.

In the source	Mathematical meaning
<code>NonUnitalCStarRepresentation.IsSingletonIrreducibleModel pi</code>	That same σ is nonzero irreducible, and every nonzero irreducible possibly nonunital $\rho : B \rightarrow B(K')$ on any complete complex Hilbert space in the comparison universe is unitarily equivalent to σ . This means a surjective complex-linear isometry $U : K \rightarrow K'$ satisfies $U\sigma(b) = \rho(b)U$ for every $b \in B$.
<code>¬ IsCompactOperatorModel pi</code>	The same σ does not simultaneously have all three properties: injectivity, compactness of every $\sigma(b)$, and coverage of every compact operator on K . It is the full compact-operator model that is excluded, not a single compact image.
<code>HasDensityCharacter A κ</code>	The same B has a norm-dense subset of cardinality $\mathfrak{c}$, and every norm-dense subset has cardinality at least $\mathfrak{c}$. This is exact norm density of the existential algebra; no faithful separable model is imposed.

Further source notes: The colon is a type annotation, and `.{0}` specifies a universe level: it is not the cardinal number zero and imposes no countability assumption. The cardinal universe convention explains this type. The separately linked `ExistsNaimarkCounterexampleOfDensity` quantifies over the algebra, Hilbert space and representations in that universe. In the linked proof, `AtomicCounterexampleAlgebra` is T , `SelectedAtomicHilbert` completedRootPureState is $\mathcal{H}$, and `.toNonUnitalStarAlgHom` retains the same inclusion ρ_0 while forgetting the requirement to preserve the identity. `inferInstance` supplies the existing C^* -algebra, Hilbert-space and spectral-order structures; these are not extra hypotheses of the theorem.

Separate complete definition of the existential result predicate, from the fixed source, lines 56–72. Its letters A, H, π denote the same B, K, σ explained in the table; this is not a continuation of the theorem signature.

```
def ExistsNaimarkCounterexampleOfDensity (κ : Cardinal.{u}) : Prop :=
  ∃ (A : Type u) (iA : NonUnitalCStarAlgebra A),
    letI := iA
  ∃ (oA : PartialOrder A),
    letI := oA
  ∃ (sA : StarOrderedRing A) (nA : Nontrivial A),
    letI := sA
    letI := nA
  ∃ (H : Type u) (nH : NormedAddCommGroup H),
    letI := nH
  ∃ (iH : InnerProductSpace ℂ H),
    letI := iH
  ∃ (cH : CompleteSpace H),
    letI := cH
  ∃ pi : NonUnitalCStarRepresentation A H,
    NonUnitalCStarRepresentation.IsSingletonIrreducibleModel.{u, u, u} pi ∧
    (¬ IsCompactOperatorModel pi) ∧ HasDensityCharacter A κ
```

▼ **Exact content identity**

Declaration: MathlibAnnex.CStarAlgebra.CAR.existsNaimarkCounterexampleOfDensity_continuum

Accepted content SHA-256: ac159c66a014ea67c887a6885485419b0ef5c4f9f437c1a8759e3fe21ef9d3fb

Accepted source guide SHA-256: db51c328deb14579b22abd0b9e9333a5fd31f62f353baa5e546103c12fd3979b

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